

Carlo Bosio

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WHO I AM

Robotics and AI researcher and engineer interested in controls, optimization, and learning methods for robot control and design automation.

EDUCATION

UC Berkeley

Expected 2027

Ph.D. in Mechanical Engineering (Controls focus)

Advisor: Mark W. Mueller

University of Pisa

2022

M.S. Robotics (ranked 1st/248)

Advisor: Mark Cutkosky (Stanford University)

Thesis: “Grasping Through Microspines: Analysis of a Three Finger Gripper for a Space Exploration Robot”

PUBLICATIONS

For a full list of publications, check my [Google Scholar](#) page.

Bosio C., Mueller M. W. “Full-autonomy co-design of collaborative air transport systems,” *International Journal of Robotics Research* (under review), 2026.

Sun Y., et al. “Agents’ Last Exam.” *arXiv preprint arXiv:2606.05405*, 2026.

Foss E. *, Tai A. *, **Bosio C.***, Mueller M. W. “Energy-efficient collaborative transport of tether-suspended payloads via rotating equilibrium.” *arXiv preprint arXiv:2603.06955*, 2026.

Jacquemont D., **Bosio C.**, Yang T., Zhang R., Orun O., Li S., Alam R., Schutzius T.M., Makiharju S.A. and Mueller M.W. “Autonomous Close-Proximity Photovoltaic Panel Coating Using a Quadcopter.” *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026.

Bosio C., and Mark W. Mueller. “Design of Autonomous Drones.” *Europhysics News* 57.3 (13-15), 2026.

Bosio C., Bosio, C., Guarrera, M., Sangiovanni-Vincentelli, A., Mueller, M. W. “Combining Large Language Models and Gradient-Free Optimization for Automatic Control Policy Synthesis.” *American Control Conference*, 2026.

Center for AI Safety agibenchmark@safe.ai “A benchmark of expert-level academic questions to assess AI capabilities.” *Nature* 649.8099: 1139-1146, 2026.

Bosio C., et al. “RDAR: Reward-Driven Agent Relevance Estimation for Autonomous Driving.” *arXiv preprint arXiv:2509.19789*, 2025.

Bosio C., Mueller M. W. “Synthesizing Interpretable Control Policies through Large Language Model Guided Search,” *American Control Conference*, 2025.

Bosio C., Mueller M. W., “Automated Layout and Control Co-Design of Robust Multi-UAV Transportation Systems,” *IEEE Robotics and Automation Letters*, 2025.

Chen T. G., Newdick S., Di J., **Bosio C.**, Ongole N., Lapôtre M., Pavone M., Cutkosky M., “Locomotion as manipulation with ReachBot,” *Science Robotics*, 2024.

EXPERIENCE

Graduate Student - UC Berkeley HiPeRLab - Berkeley AI Research Lab

2022 – Ongoing

- Research on Controls, UAVs, Design Automation, Large Language Models and Program Synthesis.
- Teaching Assistant for courses “ME136/236 – Dynamics and Control of Autonomous Flight” (150 graduate and undergraduate students), “AE10 – Control of UAVs” (50 undergraduate students). Tutor for Preliminary exam (1st year PhD students) in Controls.
- Mentored ~ 10 undergraduate and masters student researchers.
- Cross-departmental collaboration with Prof. Alberto Sangiovanni Vincentelli’s group.

Part time consultant - SWNG AI Machine Learning Software

Summer 2026

- Large scale machine learning and data analysis for chemical plant diagnostics.
- Implemented LLM-assisted search pipeline to select best failure prediction model.
- Collaborated with fast-paced team in startup environment.

Software Intern - Zoox Inc. Prediction and Behavior ML Division

Summer 2025

- Worked on estimating agent relevance in driving scenes for onboard compute prioritization.
- Set up large training pipelines using imitation and reinforcement learning.
- Enabled state-of-the-art driving performance processing an order of magnitude fewer agents.

Visiting Student Researcher - Stanford University Biomimetics and Dexterous Manipulation Lab

2022

- Research fellowship supported by NASA.
- Worked on ReachBot, project focused on a new concept for a space exploration robot. Developed a model and a simulation based on a Monte-Carlo method for the performance evaluation of a microspine gripper.
- Experimentally validated probabilistic method on real hardware.

Research Fellow - EPFL E3 Scholar CREATE Lab - Supervisor: Josie Hughes

2021
Lausanne, CH

- Developed from scratch a flexible, fully 3D printed robotic hand. I took care of the design, the grasp planning and the control of the device.
- In parallel, I also contributed to the design optimization and optimal control of a swimming soft robot.
- Collaborated with interdisciplinary team of researchers.

HONORS AND AWARDS

Qualcomm Innovation Fellowship Finalist 2025 Proposal: *Combining Large Language Models and Gradient-Free Optimization for Automatic Control Policy Synthesis.*

KTH RPL Summer School 2024 Selected as fully funded attendee for the KTH RPL Summer School in Stockholm, Sweden (acceptance rate: 3%).

Powley Fund Research Grant: Awarded 30k research funding grant.

EPFL Excellence in Engineering 2021: Awarded the highly competitive E3 Summer Research Fellowship from the EPFL School of Engineering (acceptance rate: 2.3%).

Sant’Anna School of Advanced Studies Full Scholarship 2017/22: Awarded full scholarship for undergraduate and masters studies (acceptance rate: 3%).

SKILLS

Programming Languages: C/C++, Python, Linux Shell, Matlab, R

Programming Frameworks/Libraries: ROS, Git, PyTorch, LaTeX

Spoken Languages: Italian (native), English (fluent, TOEFL iBT 111/120), French (fluent, DELF B2, EsaBac diploma), Chinese (basic)

VOLUNTEERING

ISSNAF (Italian Scientists and Scholars in North America Foundation): Organized cultural events across the San Francisco Bay Area.